DAHEJ / INDUSTRIAL IT CASE STUDY

Connected plant infrastructure.

Network, surveillance and smart-meter IT enablement

A D-TECH engagement covering structured network installation, CCTV and Wi-Fi services, plus fibre-linked connectivity for the smart-meter IT work package.

TOTAL VALUE: US$ 2,418

Including GST | INR 229,817.50 | INR 95.055 per USD
Reuters closing rate, 15 Oct 2026; indicative conversion.

03 ORDERS | 02 WORK PACKAGES

Toyo Ink India Private Limited | Dahej Plant



01 / ENGAGEMENT AT A GLANCE

Two workstreams. One infrastructure partner.

Customer: Toyo Ink India Private Limited | Dahej Phase-II, Village Jolva, Taluka Vagra, Bharuch.

Order / date	Documented scope	Basic / gross INR
6000438907 28 November 2025	PSA-II installation services: bullet cameras, 32-channel NVR, fibre, 12-port LIU, 24-port switch, Wi-Fi AP, structured cabling and 12U rack.	96,260.00 / 113,586.80
6000453065 10 February 2026	Smart-meter IT supply: 8-port PoE switch, 6-core SM fibre, Cat6, SFPs, LIUs, patch cords, racks and wiring accessories.	75,478.60 / 89,064.74
6000453066 10 February 2026	Smart-meter IT services: fibre splicing and installation / configuration.	23,022.00 / 27,165.96
Combined order value	Basic INR 194,760.60 + GST INR 35,056.90	INR 229,817.50 gross

Case study basis

Orders confirm contracted scope and quantities, not completion or measured performance. The diagrams explain the working solution concept; actual topology, configuration and acceptance require the site records.

02 / SOLUTION ARCHITECTURE

Separate the service networks and field extension

Conceptual relationship between the two ordered work packages; this is not an as-built network drawing.

PSA-II NETWORK SERVICES

Bullet camera installation
Wi-Fi access-point installation
Cat6 outlets and structured cabling

→

PSA-II AGGREGATION

24-port switch installation
24-port loaded patch panel
12U rack dressing and 12-port LIU

→

RECORDING / PLANT LAN

32-channel NVR installation
and configuration
Existing plant LAN interface to confirm

SMART-METER IT EDGE

Meter communication equipment
(customer / other-vendor interface)
305 m Cat6 ordered

⇢

FIELD NETWORK EXTENSION

1 x GWN7801P PoE switch
2 x DEM-310GT SFPs
200 m 6-core SM fibre ordered

→

UPSTREAM CONNECTION

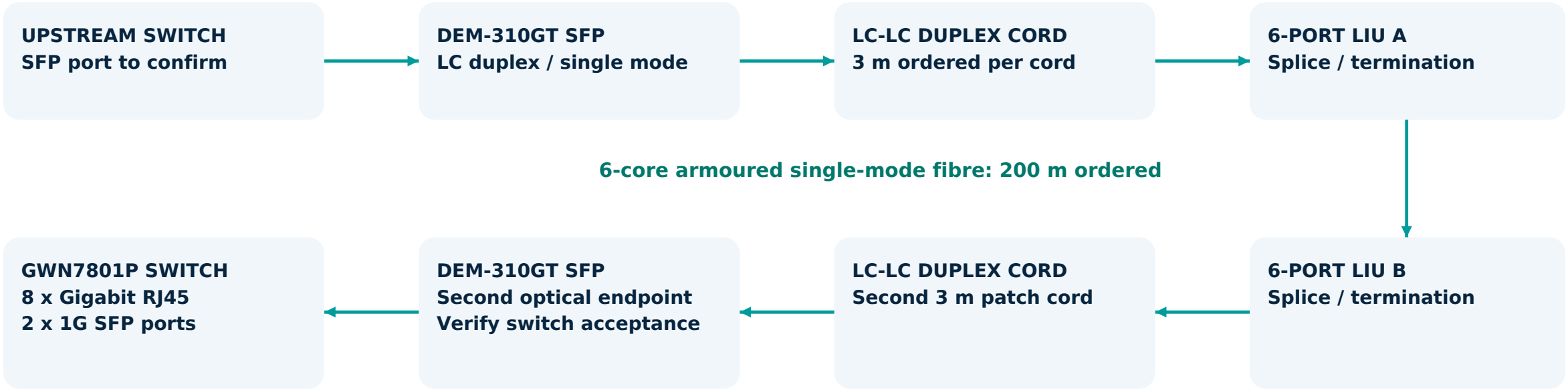
Remote SFP-capable switch / port
Application destination and VLANs
to be confirmed with plant IT

Solid: ordered functional relationship. Dashed: third-party endpoint interface. No physical link between the two workstreams is asserted.

03 / MICRO-LEVEL FIBRE DESIGN

A traceable optical path from rack to rack

Reference link arrangement using the supplied fibre components; endpoint placement remains subject to site confirmation.



One duplex optical circuit uses a fibre pair. Allocation of remaining cores and final terminations must be recorded.

Splicing service: 12 jobs in the order remarks. Validate endpoint/core schedule, polarity, insertion loss and link stability.

Engineering check: validate cross-brand SFP compatibility; matching speed and connector type alone do not prove interoperability.

04 / SURVEILLANCE AND WI-FI SERVICE DESIGN

Structured access for cameras, users and recording

PSA-II work order 6000438907 defines installation activities; equipment quantities and most models are not specified.

CAMERA SERVICE PATH

Bullet camera → Cat6 route
Outlet / connector termination
Recording connectivity test

24-PORT SWITCH

Install and label connections
Validate uplink and endpoint reachability
Port / IP mapping

32-CHANNEL NVR

Install and configure
Confirm camera mapping, recording and playback during acceptance

WI-FI AP SERVICE

Install and verify AP connectivity. Agree SSID, authentication and permitted network access with plant IT.

PASSIVE INFRASTRUCTURE

Fibre laying / splicing, 12-port LIU, Cat6 laying, 24-port loaded patch panel, IO punching, faceplates and back boxes.

RACK AND HANDOVER

Install and dress the 12U rack. Label patching and provide an agreed port / cable register for support.

A 32-channel NVR description is not a count of installed cameras. Disk capacity, retention period, analytics and AP counts are not established by these orders.

05 / SUPPLY DETAIL AND INSTALLATION CONTROLS

Materials selected for a practical field extension

Smart-meter IT supply order 6000453065 and service order 6000453066.

Ordered element	Quantity	Implementation focus
Grandstream GWN7801P	1 switch	8-port Gigabit PoE access; validate connected load and approved management settings.
Armoured single-mode OFC / Cat6	200 m / 305 m	Approve routes, segregation, bend radius, identification and termination plan.
DEM-310GT / 6-port LC LIU	2 SFPs / 2 LIUs	Document optical endpoints, test polarity / insertion loss and confirm transceiver acceptance.
LC-LC SM duplex patch cord	2 x 3 m	Clean and inspect connectors; identify ports and maintain fibre slack.
Elixir 6U IP55 / 4U rack	1 each	Confirm placement, local power, rack earthing, cable entries, heat and environmental suitability.
STP RJ45 / wiring accessories	12 connectors / 500 m line	Resolve unit / packaging wording and measure accessory consumption against the approved route.

Service control: 12 fibre-splicing jobs are described in the service remarks; installation / configuration is ordered as one job.

An IP55 enclosure does not by itself certify the installed switch or the full assembly for a hazardous area.

06 / DELIVERY METHODOLOGY

Route planning to verified handover

Recommended execution and acceptance flow for the ordered infrastructure scope.

01 Survey and agree

Verify equipment locations, cable routes, existing uplink availability and work permits. Confirm the two work-package boundaries.

02 Install passive plant

Lay fibre and Cat6, mount racks / LIUs and complete splicing, terminations, port labels and cable dressing.

03 Configure connectivity

Verify SFP link, switch management access and approved IP / VLAN plan. Confirm meter interface responsibilities.

04 Validate services

Test wired paths, AP access and NVR connectivity. Check recording / playback and endpoint reachability for the applicable scope.

05 Record evidence

Retain fibre-loss results, copper test records, port maps and configuration backups. Log deviations for closure.

06 Handover and support

Provide operator walkthrough, documents and open-item register. Obtain scope-based acceptance and agree support ownership.

07 / D-TECH VALUE AND PROJECT RECORDS

Infrastructure that is understandable and supportable

Intended value: clear workstream ownership, traceable cabling, documented interfaces and repeatable troubleshooting.

PLANT CONNECTIVITY

A fibre-based field extension with a defined switch, patching and rack pathway.

OPERATIONAL VISIBILITY

Coordinated network, Wi-Fi and CCTV installation services with recording checks.

SERVICE READINESS

Port records, test evidence and configuration documentation to aid incident isolation.

Source and validation notes

Orders: 6000438907 (28 Nov 2025), 6000453065 and 6000453066 (10 Feb 2026). The latter pair explicitly states smart-meter installation - IT part. Completion and performance records were not included.

Clarifications: supply connector line states qty 12 while remarks include “100 PC”; do not multiply automatically. The service installation line is INR 20,622, while its remark says INR 20,655; the PO line value is used in totals.

Grandstream official GWN780x quick installation guide

D-Link official DEM-310GT product specification

support.dtechindia.com

Support: +91 99989 03042

08 / FULL-PAGE SOLUTION SCHEMATIC

Plant connectivity and field-link schematic

Functional reference only. Each work package is shown separately; no inter-package connection is implied.

A | PSA-II CCTV + WI-FI SERVICES

BULLET CAMERAS

Install / terminate
Quantities not stated

WI-FI ACCESS POINTS

Install / connect
Models / counts to confirm

Cat6 / approved patching

24-PORT SWITCH

Patch panel + IO links
12U rack / fibre LIU

32-CHANNEL NVR

Configure recording / playback

B | SMART-METER IT EXTENSION

METER INTERFACE

External equipment
Not itemized in these POs

Cat6

GWN7801P SWITCH

8 x Gigabit RJ45
2 x 1G SFP

200 m 6-core SM OFC

FIELD LIU + SFP

LC duplex patch cord
One DEM-310GT endpoint

UPSTREAM LIU + SFP

LC duplex patch cord
Second DEM-310GT endpoint

UPSTREAM PLANT LAN

SFP host / destination to confirm

Solid = conceptual cable / service path. Dashed = external meter interface. Actual device placement and port configuration require approved site drawings.